MGC: MLP-Guided CNN Pretraining Using a Small-Scale Dataset for Remote Sensing Images

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***Abstract*— To overcome the inherent domain gap between natural images and remote sensing images (RSIs), it is highly desirable to develop pretraining methods specifically for RSIs. Considering the lack of widely recognized large-scale benchmarks like ImageNet in the RSI community and limited computational resources, this article proposes multilayer perceptron (MLP)-guided convolutional neural network (CNN) (MGC), a method that employs an MLP to guide the pretraining of a CNN from small-scale datasets for RSIs. MGC has two encoders, each consisting of a CNN branch and an MLP branch. We first contrast pairwise samples from the same type of branches or different types of branches across the encoders and employ a positive-pair guidance strategy to explore consistency. Due to the inherent locality issue of shallow layers in a CNN, the CNN branches often do not attend to correct foreground regions such as objects, regions of interest, and land coverage. Therefore, we further propose an attention guidance strategy to guide the CNN branches to focus on foreground regions and learn discriminative representations effectively. The proposed MGC method is validated by pretraining a CNN model using the MGC and applying it to different downstream tasks including scene classification, rotated object detection, semantic segmentation, and change detection on ten datasets. Results have confirmed the effectiveness of the proposed MGC. Our code will be released at: https://github.com/benesakitam/MGC.**

***Index Terms*— Attention, contrastive learning, pretraining, remote sensing image (RSI), self-supervised representation learning.**

I. Introduction

DEEP learning has played a major role in remote sensing image (RSI) interpretation [1], [2], [3], [4], [5], [6], [7], [8]. Many convolutional neural network (CNN)-based models, such as ResNet-50 [9], often rely on pretraining the models on ImageNet [10]. The inherent domain gap in data characteristics and imaging mechanisms between natural images and RSIs limit the performance of these models. Although some works [11], [12], [13], [14] have explored pretraining specifically on RSIs, they require diverse sources of RSIs and there is still no commonly accepted pretraining benchmark like ImageNet in the RSI community. On the other hand, pretraining on a large-scale dataset requires significant computing resources, making it only feasible in a limited number of research institutions. This article aims to generate competitive pretrained models on randomly sampled small-scale RSIs and with limited computing resources, e.g., four NVIDIA V100 GPUs.

Specifically, this article employs a contrastive learning-based CNN pretraining paradigm [16], [17], [18], [19]. While CNN often relies on locality to model discriminative features presented in images, contrastive learning mines consistency between pairwise representations. In addition, locality is more apparent in natural images rather than in RSIs probably due to low pixel resolution and nonvisible spectra bands of the latter. For instance, Fig. 1 shows the attention maps of each layer of a CNN pretrained using momentum contrastive learning [19].

For a natural scene in Fig. 1(a), layer1 and layer2 have a high probability of capturing the discrepancies between the dog and the background due to the single and large size of the foreground object dog, as well as the clear contrast with the background. As the receptive field of layer3 becomes larger, it gradually locates the discriminative regions of the dog. By layer4, the attention map includes correct foreground regions and generates high-quality discriminative representation. In contrast, remote sensing typically covers wide and complex terrains, and variation in the quality of satellite sensors may lead to blurriness or distortions in RSIs. Also, major discriminative elements such as foreground objects, regions of interest, and land cover in the image are nonuniform in scale and distribution. For instance in Fig. 1(b), CNN's layer1 and layer2 can only learn the differences between the foreground and background from limited local regions, which restricts layer3 to extract features for the correct foreground BR in larger regions. Furthermore, it results in layer4 paying attention to the background, such as the sea surface, and insufficient attention to the BR. Locality of shallow layers causes the CNN to focus on the undesired foreground regions. The

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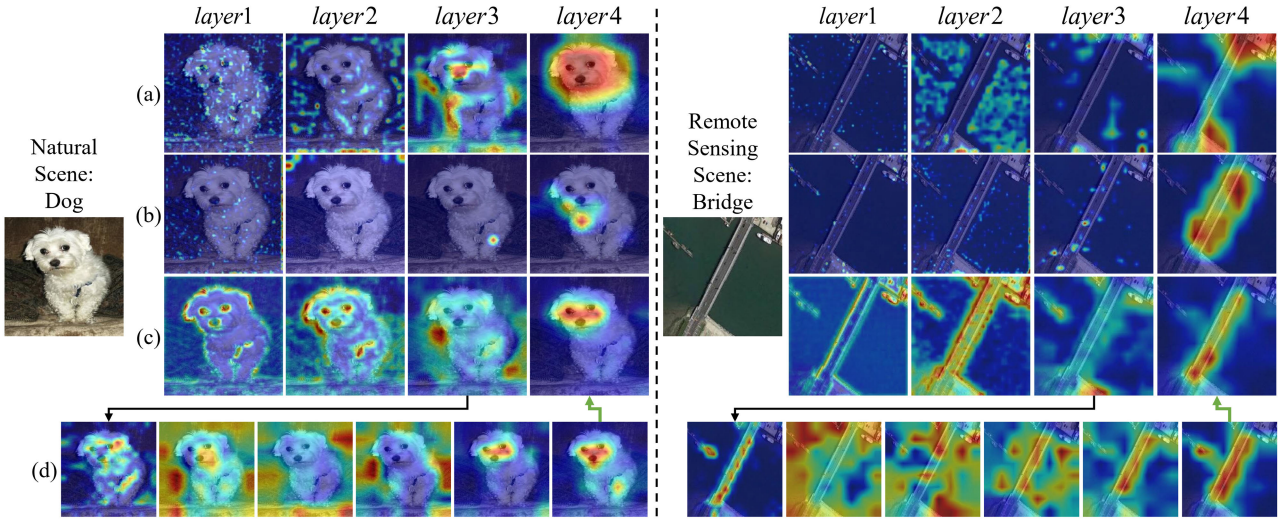


Fig. 1. Attention maps visualized by GradCAM++ [15] at each layer of different pretrained models. The warm region represents attention maps. (a) ResNet-50 pretrained on ImageNet-1k using MoCo v2. (b) ResNet-50 pretrained on small-scale RSIs using MoCo v2. The domain gap between natural scenes and remote sensing scenes has a significant negative impact on the attention of CNNs. In addition, the locality issue of shallow layers limits a CNN to pay attention to the desired foreground bridge (BR) in the remote sensing scene. (c) ResNet-50 pretrained on small-scale RSIs using the proposed MGC. (d) MLP branch pretrained on small-scale RSIs using MGC. The MLP branch gradually locates the complete BR by aggregating spatial context from diverse regions, thereby guiding the CNN to attend the correct foreground BR.

low-quality representation extracted from it is not conducive to exploring consistency between samples. Therefore, we propose an MLP-guided CNN (MGC) pretraining method for RSIs. MGC introduces an additional multilayer perceptron (MLP) branch to overcome the locality issue of shallow layers and guide the CNN branch to focus on correct foreground regions. The MLP branch consists of three convolutional layers and multiple MLPs that aggregate spatial contextual information from different positions in the feature map. In the remote sensing scene of Fig. 1(c) and (d), despite layer3 incorrectly focusing on the land cover at the lower right corner of the BR, each layer of the MLP branch gradually locates a complete BR by aggregating features from more diverse regions, thus guiding the CNN to attend the correct foreground regions.

Specifically, MGC follows a classic contrastive pretraining paradigm with a Siamese encoder architecture. Each encoder consists of a CNN branch and an MLP branch. The MLP branch focuses on foreground regions and mines relevant region for discriminative representation. MGC employs same-branch and cross-branch contrastive learning to enhance consistency by contrasting paired representations from the same type of branches (CNN branches or MLP branches of two encoders) or different types of branches (the CNN branch of one encoder and the MLP branch of another encoder). Meanwhile, utilizing stable features created by the momentum update [16], [18], a positive-pair guidance strategy is developed to improve contrastive learning within the same branches. Furthermore, an attention guidance strategy is proposed to overcome the locality issue of shallow layers in the CNN branches with the help of the region discriminative information mined by the MLP branches, enforcing the CNN branches to gradually pay attention to correct foreground regions.

Using only 61 272 RSIs randomly sampled from the Aerial Image Dataset (AID) [20], MLRSNet [21], and NWPU-RESISC45 (NWPU45) [22] datasets, MGC can learn discriminative and consistent representation. By adjusting the range of aggregating contexts in the MLP branches, two versions of pretrained models, MGC-Base and MGC-Large, are generated. MGC-B outperformed all compared models in the downstream classification task. Moreover, competitive performance was also achieved on other predictive tasks. For instance, MGC-B improved mean average precision (mAP) by 0.6 compared to Oriented R-CNN [23] detector in rotated object detection on DOTA [24]. In semantic segmentation, MGC-L outperforms Deeplabv3+ [25] by 1.08 mIoU on iSAID [26]. In change detection, F1-score of MGC-B on LEVIR [27] is 0.25 better than MoCo v2.

The contributions of this article can be summarized as follows.

1) MGC, an MLP-guided CNN pretraining method using only small-scale RSIs, is proposed. MGC is capable of guiding CNNs to overcome the locality issue of shallow layers and focus on the desired foreground regions via the proposed MLPs.
2) A representation learning framework that incorporates the same-branch and cross-branch contrastive learning, along with the assistance of a positive-pair strategy, is developed to improve consistency.
3) An attention guidance strategy is proposed to enforce the CNNs to pay more attention to the correct foreground regions, hence, to learn effective representation.
4) Models pretrained using MGC are extensively evaluated on ten datasets for different downstream tasks such as scene classification, rotated object detection, semantic segmentation, and change detection. The results have demonstrated the superior performance of MGC.

II. Related Work

A. Contrastive Learning

Contrastive learning explores consistent visual representation by pulling the representation of positive pairs (i.e., views

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of the same image) closer while pushing negative pairs (i.e., views of different images) farther away. SimCLR [17] is a simple contrastive framework in which learn consistency from the augmented views of other negative samples in a minibatch. MoCo [16] uses a momentum-updated encoder to improve representation learning. BYOL [18] and its variant SimSiam [28] use positive pairs only to mine mutual information. SwAV [29] clusters samples instead of computing pairwise representations. In addition, varieties of contrastive methods, such as patch-level representations contrasting [30], hard sample mining [31], [32], prototype selecting [33], [34], mutual information increasing [35], [36], transformer bootstrapping [37], relationship alignment [38], and supervised contrasting [39], are continuously improving pretraining for natural images.

B. Remote Sensing Pretraining

Pretraining methods for RSIs generally follow the strategies used for nature images and can be broadly divided into two basic paradigms, both of which are self-supervised contrastive learning [16], [17], [18] and masked image modeling [40], [41].

In contrastive learning methods [12], [42], [43], [44], CNNs were commonly used as a backbone network. These methods leverage the inherent properties of RSIs to satisfy the requirement of learning unique consistency in a latent space between pairwise samples. SeCo [12] learns seasonal invariance between positive pairs. Ayush et al. [43] employed temporal information and geographical location to design pairwise samples. MATTER [42] leverages multitemporal, spatially aligned RSIs to learn material and texture representation consistency. GeoKR [44] used geographical knowledge to provide supervision and a spatially augmented momentum contrastive framework was proposed in [45].

The global self-attention mechanism of vision transformer [46] allows modeling relationship among masked and visible regions for accurate pixel prediction, which has made masked image modeling methods popular in self-supervised representation learning. However, it is worth noting that vision transformer-based [46] methods often require huge computational resources. RingMo [14] collected two million images from both satellite and aerial platforms, and designs a patch incomplete mask strategy to recover the masked RSI according to the properties of scenes and objects. SatMAE [13] incorporated additional temporal and multispectral information in the positional encoding to capture the spatiotemporal relationships in the expanded functional map of the world (fMoW) dataset [47]. By leveraging the absolute ground sample distance information, Scale-MAE [48] was able to reconstruct masked RSI at different resolutions.

In addition, RSP [11] fully studied supervised pretraining on the MillionAID dataset [49], which is similar in scale to ImageNet-1k. The above methods all require large-scale datasets from diverse sources and demand high computational resource for pretraining. In this article, the proposed MGC neither relies on specific RSI properties, such as temporal and geographical information nor large-scale datasets. MGC improves the baseline CNNs to learn consistent and discriminative representations of correct foreground regions by introducing MLP paths in a Siamese architecture.

III. Method

A. Dual-Branch Encoder

MGC adopts the classic Siamese encoder architecture with a contrastive pretraining paradigm. The encoder k is a momentum update [16], [18] of the encoder q, i.e., $\theta_k \leftarrow \alpha \cdot \theta_k + (1-\alpha)\cdot\theta_q$, where θ_q and θ_k mean the encoder parameters. Each encoder in MGC, as depicted in Fig. 2(a), consists of a regular CNN branch (E_q^c/E_k^c) and a foreground-sensitive MLP branch (E_q^m/E_k^m). Following the same random data augmentation as in [17], two views of an input RSI x, v_q and v_k, are created and subsequently fed into each branch of the two encoders, thereby generating diverse representations. The grayed-out c_{k^-} and m_{k^-} in Fig. 2(a) denote features extracted from other images excluding x.

1) CNN Branch: CNN branches $E_\cdot^c$ include E_q^c from encoder q and E_k^c from the momentum encoder k, all consisting of the backbone ResNet-50 connected with a projector [17], and E_q^c links an extra predictor [18]. The projector is the same as the predictor and consists of a Channel fully connected (FC) layer (i.e., linear layer), BatchNorm, rectified linear unit (ReLU), and another Channel FC layer.

2) MLP Branch: The only difference between MLP branches $E_\cdot^m$ and $E_\cdot^c$ lies in the backbone structure. In $E_\cdot^m$'s backbone, we retain the first three layers of ResNet-50 and connect with an MLP network, as shown in Fig. 3, consisting of two general MLPs and n CycleMLPs [50]. The Cycle FC layer in CycleMLP can shift and sample along the channel of the feature map circularly and aggregate spatial contextual information in height and width dimensions; it can be calculated as follows:

$$M_{h,w,:}^{\text{cycleFC}} = \sum_{c=0}^{c_{\text{in}}} W_c \cdot F_{h+\Delta h, w+\Delta w, c} + b \tag{1}$$

where $F_{h+\Delta h,w+\Delta w,c}$ denotes the values of cth channel at the dynamic cycle positions $(h+\Delta h, w+\Delta w)$. $W_c \in \mathbb{R}^{C_{\text{in}}\times C_{\text{out}}}$ and $b \in \mathbb{R}^{C_{\text{out}}}$ are, respectively, the weight and bias of Cycle FC. The offsets of height and width in cth channel are obtained by $\Delta h = (c \bmod s) - 1$ and $\Delta w = (\lfloor c/s \rfloor \bmod s) - 1$, respectively, where s is an adjustable step size.

A larger step size s means that $E_\cdot^m$ aggregates more spatial contextual information. Therefore, as the training continues, $E_\cdot^m$ can mine discriminative representations associated with foreground regions. With the increase in s, $E_\cdot^m$ encompasses larger and discriminative foreground regions.

B. Workflow

The pipeline of each proposed component in MGC is shown in Fig. 2. First, the same-branch contrastive learning utilizes features c_q extracted by the CNN branch E_q^c of encoder q and c_k from E_k^c of the momentum encoder k, or features m_q and m_k provided by the MLPs $E_\cdot^m$, to construct positive and negative pairs, employing the conventional contrastive learning [16], [17], [19] to explore consistency. Meanwhile, the positive-pair guidance strategy drives c_q (m_q) to learn information from its

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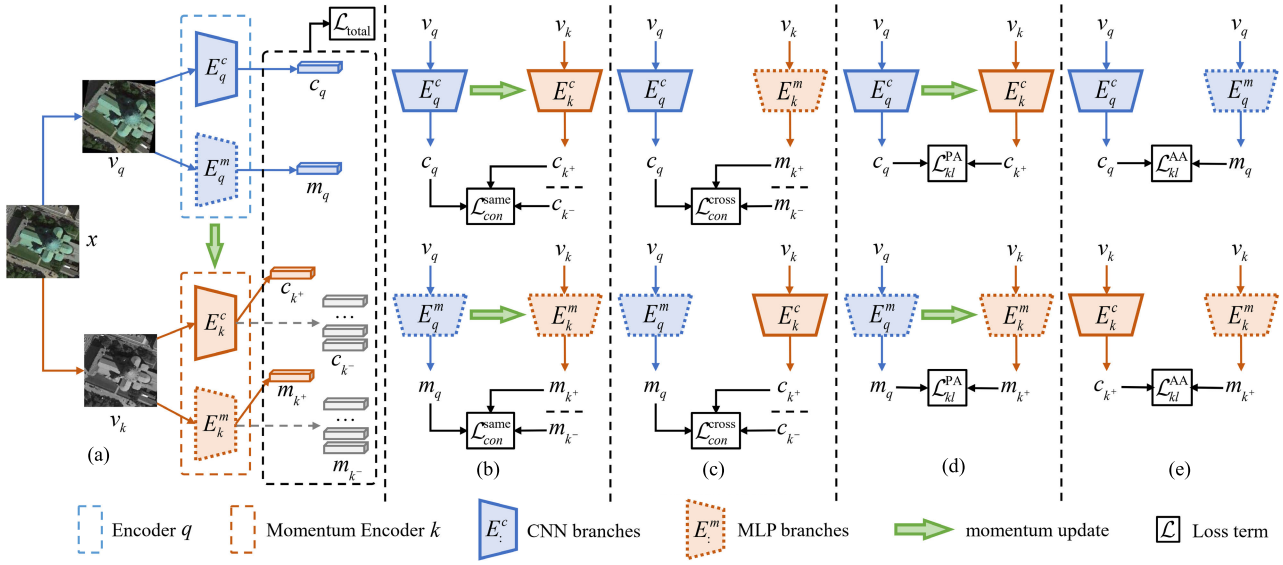


Fig. 2. (a) MGC's pipeline. By minimizing the total loss L_{total} that integrates the four proposed components of (b)–(e), MGC can generate the competitive pretrained model. (b) Enables the model to obtain the fundamental consistent information. (c) Help the model to further learn the consistency by exploiting between representations from distinct branches. (d) Improves the same-branch contrastive learning based on stable features provided by the momentum encoder k. (e) Guides $E_{:}^{c}$ to pay attention to the correct foreground regions and extracts discriminative representations.

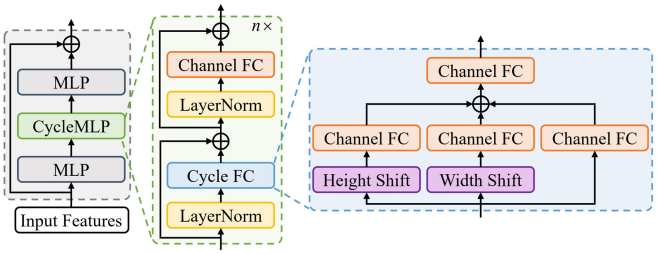


Fig. 3. MLP network consists of two general MLPs and n CycleMLPs in a residual connection manner.

positive sample c_{k^+} (m_{k^+}), thus facilitating the same-branch contrastive learning. Next, cross-branch contrastive learning selects c_q from E_q^c and m_{k^+} and m_{k^-} from E_k^m to construct paired samples; similarly, m_q is also utilized to create positive and negative pairs with c_{k^+} and c_{k^-}, further enriching overall consistency. Finally, the proposed attention guidance strategy fits the distribution of features from $E_{:}^{c}$ and $E_{:}^{m}$ within the same encoder. This drives $E_{:}^{m}$ to guide $E_{:}^{c}$ to focus on foreground regions and learn discriminative representations, thereby overcoming the locality issue of shallow layers of $E_{:}^{c}$.

C. Same-Branch Contrastive Learning

Like [16], [17], and [19], the goal of the same-branch contrastive learning is to retrieve the relevant "keys" for a "query" and make them close to each other in the latent space, which is achieved by minimizing the infoNCE loss [51]. After obtaining c_q and c_k encoded by $E_{:}^{c}$, the loss is calculated as

$$\mathcal{L}_{c\leftrightarrow c}(c_q, c_k) = -\log \frac{\exp(c_q \cdot c_{k^+}/\tau)}{\exp(c_q \cdot c_{k^+}/\tau) + \sum_{k_-} \exp(c_q \cdot c_{k^-}/\tau)} \tag{2}$$

where τ is a temperature parameter [52], and c_{k^+} and c_q are from different views of the same image, i.e., a positive pair. c_{k^-} is extracted from other images and is often considered as a negative sample of c_q. Certainly, m_q and m_k from $E_{:}^{m}$ have the same settings to compute the infoNCE loss between MLP branches. The same-branch contrastive learning motivates the model to learn the fundamental consistent information. The full loss is written as

$$\mathcal{L}_{\text{con}}^{\text{same}} = \mathcal{L}_{c\leftrightarrow c}(c_q, c_k) + \mathcal{L}_{m\leftrightarrow m}(m_q, m_k). \tag{3}$$

D. Cross-Branch Contrastive Learning

$E_{:}^{c}$ extracts local features from any regions in the scene. The aggregated spatial context of the larger scale scene allows $E_{:}^{m}$ to focus on foreground regions, enabling it to learn region discriminative representations that are sensitive to the foreground. Therefore, we propose cross-branch contrastive learning to explore the mutual information from these two different representations, further improving the consistency captured by the model. For a random input image, we treat c_q extracted by E_q^c as a "query," m_{k^+} of the data augmented view encoded by E_k^m as the positive sample of c_q, and representations from other images as negative samples m_{k^-} of c_q. Similarly, when m_q is a "query," the samples used to form positive and negative pairs all come from E_k^c. The projector and predictor enable the generated feature vectors to have the same size. Constructing positive and negative pairs across branches, we also minimize the infoNCE loss as defined in (2). Therefore, the complete loss for cross-branch contrastive learning can be written as

$$\mathcal{L}_{\text{con}}^{\text{cross}} = \mathcal{L}_{c\leftrightarrow m}(c_q, m_k) + \mathcal{L}_{m\leftrightarrow c}(m_q, c_k). \tag{4}$$

E. Positive-Pair Guidance

In Fig. 2(d), due to the typically set large momentum update coefficient α [16], [18], the network parameters of two encoders are similar. The discriminative information contained

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in the features extracted by the momentum encoder k is fully preserved during each iteration, resulting in high stability. When the model contrasts a positive pair constructed from a "query" c_q and the stable feature c_{k+}, it concentrates on exploring consistency. Otherwise, the model might learn some noise between c_q and the unstable c_{k+}. Therefore, we take advantage of the momentum update and design a positive-pair guidance strategy to strengthen the same-branch contrastive learning. For $E_{\cdot}^{c}$, we minimize the Kullback-Leibler (KL) divergence between the c_q and c_{k^+} distributions

$$\mathcal{L}_{k^+\to q}^{c}\left(c_q, c_{k^+}\right) = -c_{k^+} \log \frac{c_{k^+}}{c_q}. \tag{5}$$

The above equation enforces the consistency between c_{k^+} and c_q, further promoting contrastive learning within the same type of branch. With the momentum update, E_k^c gradually has the same ability to extract representative features. The same strategy is adopted on m_q and m_{k^+} extracted from $E_{\cdot}^{m}$, and yields the following full loss function:

$$\mathcal{L}_{kl}^{\mathrm{PG}} = \mathcal{L}_{k^+\to q}^{c}\left(c_q, c_{k^+}\right) + \mathcal{L}_{k^+\to q}^{m}\left(m_q, m_{k^+}\right). \tag{6}$$

F. Attention Guidance

The locality issue of shallow layers in the CNN causes the attention of $E_{\cdot}^{c}$ inevitably includes redundant background and misses some foreground. In contrast, $E_{\cdot}^{m}$ aggregates spatial contextual information from diverse regions, driving its attention to encompass correct foreground regions. The representations learned by $E_{\cdot}^{m}$ bear sufficient discriminative power in the locations corresponding to foreground regions. This is achieved by minimizing the KL divergence between distributions of m_q and c_q

$$\mathcal{L}_{m\to c}^{q}\left(c_q, m_q\right) = -m_q \log \frac{m_q}{c_q} \tag{7}$$

where the region discriminative representation m_q acts as a supervised signal to enforce E_q^c to focus on the foreground regions in the same scene. Meanwhile, E_k^c is guided to locate the same foreground in another view. The final loss can be written as

$$\mathcal{L}_{kl}^{\mathrm{AG}} = \mathcal{L}_{m\to c}^{q}\left(c_q, m_q\right) + \mathcal{L}_{m\to c}^{k^+}(c_{k^+}, m_{k^+}). \tag{8}$$

Fig. 1 demonstrates the effectiveness of improving attention to foreground regions by (8).

G. Total Loss

E_q^c, E_q^m, E_k^c, and E_k^m are jointly optimized to perform high-quality representation learning as follows:

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{con}}^{\mathrm{same}} + \lambda_C \cdot \mathcal{L}_{\mathrm{con}}^{\mathrm{cross}} + \lambda_P \cdot \mathcal{L}_{kl}^{\mathrm{PG}} + \lambda_A \cdot \mathcal{L}_{kl}^{\mathrm{AG}} \tag{9}$$

where weight λ_C balances the cross-branch contrastive learning during model optimization, λ_P relates to positive pairs guidance loss, and λ_A controls the strength of $E_{\cdot}^{m}$ in guiding $E_{\cdot}^{c}$ to pay attention to the correct foreground regions.

IV. Experiments

To validate the proposed MGC pretraining method, ResNet-50 is used as the CNN baseline in the two encoders and the pretrained ResNet-50 in E_q^c is transferred for scene classification, rotated object detection, semantic segmentation, and change detection. Results are reported and analyzed in this section.

A. Pretraining Settings

1) Datasets: Three scene classification datasets MLRSNet [21], NWPU45 [22], and AID [20] are used to construct a pretraining dataset.

1) AID has images from different countries on Google Earth. These images are extracted at different times and seasons under different imaging conditions. The dataset contains 10 000 images with 600 × 600 pixels in 30 classes. The spatial resolution of each image is from 0.2 to 0.8 m.
2) NWPU45 is a common RSI benchmark collected by Northwestern Polytechnical University, Xi'an, China, from Google Earth. It contains 31 500 images in RGB color space, divided on average into 45 classes, with each class including 700 images. The spatial resolution of these images ranges from 0.2 to 30 m, and they are all sized at 256 × 256 pixels.
3) MLRSNet is a large-scale multilabel RSI dataset, consisting of 109 161 images collected from Google Earth. These images have spatial resolutions ranging from 0.1 to 10 m with 256 × 256 pixels. All images are assigned to 46 broad classes, and each board class contains from 1500 to 3000 images. Every image has different subtags varying from 1 to 13. In this article, only the broad class labels are used.

In particular, samples of each class in the MLRSNet and NWPU45 datasets are randomly divided into 40% for training or fine-tuning and 60% for testing. For AID, the division is 50% versus 50%. According to classes, three training sets are merged to construct a Fusion dataset containing 61 272 images of 60 classes for pretraining a CNN model. Random sampling ensures a reasonable and balanced distribution of the pretraining data, allowing us to focus on exploring an efficient pretraining paradigm without being biased by uneven data distributions. The classes contained in the Fusion dataset are listed in Table I.

2) Pretraining Details: All images from the Fusion dataset are first resized to 224 × 224 by random cropping, and new views are produced by the same data augmentation operations as [17]. In our implementation, symmetrized loss [18], [28], [29] is adopted. The model is trained with an stochastic gradient descent (SGD) optimizer, and its weight decay and momentum are 0.0001 and 0.9, respectively. The initial learning rate is set at 0.035, and the warm cosine decay schedule [17] is used. We follow the cosine design [37] to gradually increase the momentum update coefficient α from 0.99 to 1 along with training. Batchsize is 512, and the number of epochs is 800. For hyperparameters, the numbers of CycleMLP $n = 6$, weights of losses $\lambda_C = 1$, $\lambda_P = 200$,

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TABLE I
LIST OF CLASSES FOR THE PRETRAINING DATASET FUSION

Airplane, Airport, Bareland, Baseball diamond, Basketball court, Beach, Bridge, Center, Chaparral, Church, Circular farmland, Cloud, Commercial area, Dense residential area, Desert, Eroded farmland, Farmland, Forest, Freeway, Golf course, Ground track field, Harbor & Port, Industrial area, Intersection, Island, Lake, Meadow, Medium residential, Mobile home park, Mountain, Overpass, Palace, Park, Parking lot, Parkway, Railway, Railway station, Resort, Rectangular farmland, River, Roundabout, Runway, School, Sea ice, Ship, Shipping yard, Snowberg, Sparse Residential area,Square, Stadium, Storage tank, Swimming pool, Tennis court, Terrace, Thermal power station, Transmission tower, Vegetable greenhouse, Viaduct, Wetland, Wind turbine

and $\lambda_A = 50$, respectively. Based on the above parameters, two models with different step sizes are trained, i.e., MGC-Base with $s = 3$ and MGC-Large with $s = 11$. The models are implemented using PyTorch and trained on four NVIDIA Tesla V100 GPUs.

B. Scene Classification

1) Experimental Setup: In addition to the three datasets described in Section IV-A, we also conduct scene classification experiments on the EuroSAT dataset [53] to further validate the performance of the pretrained models under varying data distributions.

EuroSAT spans a land cover of 34 European countries and contains a total of 27 000 images collected from the Sentinel-2 satellite in ten classes, each consisting of 2000–3000 images with 64 × 64 pixels. The same split as proposed in [54] for training and testing is adopted.

Following the linear probe protocol [16], we freeze all layers of the pretrained model and add a supervised linear classifier to create a simple classification network. We train the network for 80 epochs and report top-1 accuracy on the testing set of AID, NWPU45, MLRSNet, and EuroSAT. In training the classifier, standard cross-entropy loss and the SGD optimizer with a momentum of 0.9 are used. In addition, the initial learning rate is 0.2, and batchsize is 256. All experiments are conducted on four V100 GPUs.

2) Comparison With SOTA: Seven state-of-the-art contrastive learning-based methods, including classic MoCo v2 [19], SimCLR [17], and BYOL [18], cluster-based SwAV [29] and HCSC [34], patch contrast-based DetCo [30], and two stream-based CARE [37], are adopted to pretrain models on Fusion for comparison. The batchsize remains 512 due to GPU memory constraint and other hyperparameters are set to their defaults. The number of epochs is set to 800 for all methods. In addition, we also compare the MGC models with other available pretrained models such as INP (traditional supervised pretrained on ImageNet [10]); RSP [11] (supervised pretrained on about 950k RSIs); SeCo [12] (seasonal contrastive pretrained on 1M RSIs); and MoCo v2 and SwAV (pretrained on ImageNet-1k for 800 epochs). All classification results are reported in Table II, where the **bold**, red, and blue mean top three scores of the metric, respectively. MGC-B achieves the best accuracy on all datasets, but MGC-L performs worse than SwAV and CARE on MLRSNet, and not as good as CARE on NWPU45 either.

TABLE II
COMPARISON OF THE SCENE CLASSIFICATION RESULTS ON THREE RSI DATASETS. † MEANS THE MGC MODEL TRAINS ONLY 200 EPOCHS

Methods	MLRSNet	NWPU45	AID	EuroSAT
	Accuracy			
Available Pre-trained Models				
INP [10]	90.97	86.79	91.08	93.57
RSP [11]	90.95	85.66	92.38	91.19
MoCo v2 [19]	91.99	88.16	92.88	95.30
SwAV [29]	93.40	90.08	93.58	95.70
SeCo [12]	71.02	69.02	72.86	67.31
Pre-trained on Fusion				
MoCo v2 [19]	94.81	92.02	85.58	91.28
SimCLR [17]	60.97	46.13	57.27	73.59
BYOL [18]	63.78	45.07	56.85	76.54
DetCo [30]	77.26	61.35	69.39	82.31
SwAV [29]	96.08	91.94	94.02	94.93
HCSC [34]	92.81	87.89	91.12	94.48
CARE [37]	96.05	92.31	93.42	95.91
MGC-B †	95.95	92.50	94.14	95.83
MGC-B	**96.30**	**92.91**	**94.72**	**96.41**
MGC-L †	95.83	92.15	94.32	95.28
MGC-L	95.68	92.07	93.84	96.11

This is because the classification task relies more on the local recognition of $E_{:}^{c}$ than the locating ability. The large s enables $E_{:}^{m}$ to aggregate a larger range of spatial contextual information, enhancing the locating ability of $E_{:}^{c}$ in large-scale scenes. In MGC-B, information within small-scale scenes learned by $E_{:}^{c}$ improves local recognition.

C. Rotated Object Detection

1) Experimental Setup: We evaluate the detection performance of the MGC pretrained models on the DOTA [24] and HRSC2016 [57] datasets using three rotated object detectors, all based on ResNet-50 backbone: Oriented R-CNN [23], RoI Transformer [55], and S^2ANet [56]. Oriented R-CNN and RoI Transformer are two-stage detectors, while S^2ANet is a one-stage detector.

1) DOTA is a large-scale rotated object detection dataset. It contains 2806 images ranging from 800 × 800 to 4000 × 4000 pixels and has 188 282 instances with rotated bounding box annotations belonging to 15 object classes, i.e., BR, Harbor (HA), Ship (SH), Plane (PL), Helicopter (HC), Small vehicle (SV), Large vehicle (LV), Baseball diamond (BD), Ground track field (GTF), Tennis court (TC), Basketball court (BC), Soccer-ball field (SBF), Roundabout (RA), Swimming pool (SP), and Storage tank (ST). In our experiments, each image is cropped to 1024 × 1024 patches with a stride of 824 and both training and validation sets are used for training.
2) HRSC2016 is a popular dataset widely used to detect rotated SHs. It has only one class and contains 1061 images ranging in size from 300 × 300 to 1500 × 900 pixels. The 436 training images and 181 validation images are used for training, and the remaining 144 images are used for testing. We fix the image aspect ratio, and the shorter edge of the image is resized to 800, while the longer edge does not exceed 1333.

All layers of the pretrained models are fine-tuned in an end-to-end manner, and the fine-tuned models are used as

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TABLE III

DETECTION RESULTS OF DIFFERENT PRETRAINED MODELS ON DOTA USING THE TWO-STAGE DETECTOR ORIENTED R-CNN [23]

Method	mAP	AP per class														
		PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC
Available Pre-trained Models																
INP [10]	75.66	89.32	82.71	**54.44**	70.86	78.82	82.95	88.13	90.88	85.82	85.30	**62.59**	62.71	74.63	69.28	56.46
RSP [11]	**76.95** (+1.29)	**89.81**	**83.60**	54.25	72.55	78.44	**83.29**	88.27	**90.91**	**86.70**	85.67	62.19	68.98	73.46	**71.58**	**64.55**
MoCo v2 [19]	72.48 (-3.18)	89.08	77.33	53.51	69.51	78.60	81.67	88.00	90.90	83.91	85.08	52.85	59.22	71.46	63.65	42.48
SwAV [29]	75.07 (-0.59)	89.59	81.40	52.59	73.67	76.99	81.89	88.13	90.90	84.80	85.36	54.65	64.82	70.28	67.81	62.83
SeCo [12]	69.95 (-5.71)	88.90	72.45	49.90	66.08	76.03	79.82	86.40	90.87	80.25	82.53	40.65	56.87	66.57	64.57	47.33
Pre-trained on Fusion																
MoCo v2 [19]	74.31 (-1.35)	89.80	82.77	52.52	69.69	77.84	81.65	87.88	90.86	85.17	83.80	55.28	60.79	73.23	69.38	53.96
SimCLR [17]	67.37 (-8.29)	88.60	71.78	47.39	64.19	73.46	76.18	85.76	90.79	76.01	80.83	38.34	53.24	66.87	61.25	35.83
BYOL [18]	48.95 (-26.71)	79.38	52.43	24.04	28.04	65.17	62.86	69.36	90.66	61.26	69.05	15.16	33.41	35.65	39.87	7.96
DetCo [30]	68.06 (-7.60)	88.72	70.53	49.49	64.65	76.64	80.14	86.40	90.86	77.94	76.30	36.79	57.54	66.61	63.11	35.24
SwAV [29]	71.85 (-3.81)	89.09	76.15	48.64	67.76	76.54	80.63	87.07	90.89	83.03	83.46	45.55	62.53	65.07	66.99	54.27
HCSC [34]	73.39 (-2.27)	89.32	81.28	53.12	66.16	77.49	80.20	87.55	90.88	82.12	84.86	52.15	65.43	71.82	68.47	50.01
CARE [37]	75.15 (-0.51)	89.39	81.52	53.20	73.40	**78.83**	82.35	87.89	90.89	85.17	**85.83**	58.65	62.98	73.76	69.46	53.98
MGC-B	76.26 (+0.60)	89.39	83.52	54.41	**75.80**	78.45	82.09	88.18	90.89	85.51	85.76	59.70	66.78	74.60	70.85	58.01
MGC-L	75.64 (-0.02)	89.69	82.12	53.83	71.11	78.12	81.40	**88.31**	90.90	84.74	85.66	56.79	**69.64**	**74.79**	71.25	56.32

TABLE IV

DETECTION RESULTS OF DIFFERENT PRETRAINED MODELS ON DOTA USING THE TWO-STAGE DETECTOR ROI TRANSFORMER [55]

Method	mAP	AP per class														
		PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC
Available Pre-trained Models																
INP [10]	76.79	89.15	83.37	**54.80**	75.46	78.54	**82.04**	87.69	90.88	86.79	85.29	64.17	67.16	75.47	72.71	58.40
RSP [11]	**78.38** (+1.59)	89.25	84.98	54.43	**78.86**	**78.68**	81.84	87.46	90.86	87.15	**86.86**	**67.66**	65.43	**75.68**	**79.22**	67.25
MoCo v2 [19]	75.07 (-1.72)	**89.60**	81.40	52.59	73.67	76.99	81.29	**88.13**	90.86	84.80	85.36	54.65	64.82	70.28	67.81	62.86
SwAV [29]	76.89 (+0.10)	89.37	**85.24**	52.48	77.85	77.12	80.95	87.72	90.87	86.24	86.02	62.18	65.23	74.25	70.15	**67.69**
SeCo [12]	72.42 (-4.37)	88.56	80.35	49.50	70.58	75.98	79.37	86.20	90.82	82.35	83.36	54.97	57.34	68.78	68.72	50.51
Pre-trained on Fusion																
MoCo v2 [19]	74.86 (-1.93)	89.36	83.22	52.10	70.79	77.22	80.49	86.71	90.79	**87.53**	84.70	63.65	61.60	74.52	71.40	48.88
SimCLR [17]	68.99 (-7.80)	87.67	73.59	45.95	66.42	73.22	74.96	83.60	90.85	79.47	81.71	46.20	54.35	67.68	65.41	43.75
BYOL [18]	66.24 (-10.55)	87.92	74.61	41.48	54.92	71.32	72.41	83.01	90.87	79.89	80.44	42.95	53.72	59.39	56.41	44.29
DetCo [30]	70.11 (-6.68)	88.40	74.97	49.53	70.83	76.06	75.43	85.04	90.85	78.91	75.30	43.03	57.00	71.48	66.25	48.59
SwAV [29]	74.78 (-2.01)	89.11	80.43	50.54	73.42	76.19	79.73	87.18	90.85	86.12	85.23	58.81	63.76	67.94	72.09	60.36
HCSC [34]	74.09 (-2.70)	89.54	82.70	53.00	68.59	73.52	76.57	86.69	90.82	82.83	85.07	57.06	64.63	73.92	72.08	54.40
CARE [37]	75.84 (-0.95)	89.17	83.08	52.42	74.93	77.55	81.62	87.26	90.86	87.38	84.90	61.69	64.81	74.37	70.90	56.60
MGC-B	76.43 (-0.36)	89.52	83.90	53.64	71.92	77.98	81.25	87.26	**90.89**	86.89	86.10	62.29	**68.19**	75.60	72.66	58.40
MGC-L	76.03 (-0.76)	89.10	83.87	53.96	78.11	74.01	77.80	87.51	90.87	80.80	86.30	63.89	66.01	75.45	74.10	58.73

TABLE V

DETECTION RESULTS OF DIFFERENT PRETRAINED MODELS ON DOTA USING THE ONE-STAGE DETECTOR S^2ANET [56]

Method	mAP	AP per class														
		PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC
Available Pre-trained Models																
INP [10]	73.88	88.41	78.84	51.23	72.89	**78.94**	79.22	87.35	90.88	85.56	84.27	**65.22**	64.26	**69.60**	67.38	44.13
RSP [11]	**75.40** (+1.52)	88.65	**82.58**	52.93	**76.26**	78.63	79.68	87.42	**90.90**	**86.49**	**86.00**	60.74	67.60	68.75	**71.11**	**53.18**
MoCo v2 [19]	67.62 (-6.26)	88.23	69.38	47.05	65.02	77.46	76.19	86.28	90.88	80.57	82.26	45.60	59.61	63.43	55.00	27.33
SwAV [29]	67.26 (-6.62)	88.60	70.34	40.11	58.00	77.89	75.90	87.11	90.85	72.89	84.43	42.34	60.21	55.97	62.41	41.88
SeCo [12]	64.87 (-9.01)	87.68	64.42	48.46	57.70	75.90	74.16	84.40	90.78	68.12	79.70	31.78	57.65	61.34	62.13	28.86
Pre-trained on Fusion																
MoCo v2 [19]	72.68 (-1.20)	88.87	77.87	50.04	68.18	78.03	77.14	86.99	90.89	86.04	84.58	54.79	66.31	65.72	67.12	47.61
SimCLR [17]	62.83 (-11.05)	87.43	64.39	43.77	55.73	74.78	72.11	82.70	90.79	71.48	78.39	27.24	54.28	59.93	55.45	23.96
BYOL [18]	62.71 (-11.17)	83.97	68.36	42.88	50.46	74.62	71.65	81.47	90.82	73.03	75.22	36.01	56.37	52.29	50.77	32.72
DetCo [30]	64.32 (-9.56)	87.48	64.99	46.34	60.98	75.95	74.18	83.08	90.87	68.3	73.31	33.88	56.07	62.02	58.57	28.73
SwAV [29]	69.48 (-4.4)	88.79	75.86	48.19	48.58	77.19	75.22	86.60	90.88	81.10	83.39	51.22	64.39	58.89	67.21	44.64
HCSC [34]	71.77 (-2.11)	88.70	77.79	51.22	67.59	77.56	76.56	86.10	90.86	81.56	83.50	52.88	64.49	65.11	68.22	44.35
CARE [37]	71.62 (-2.26)	88.67	75.35	50.20	71.58	78.12	77.21	86.71	90.88	80.66	84.07	54.05	64.70	67.50	61.24	43.29
MGC-B	73.87 (-0.01)	**88.93**	78.65	52.27	73.43	78.08	79.80	87.18	90.86	81.94	84.98	58.64	66.02	66.47	70.37	50.47
MGC-L	74.03 (+0.15)	88.45	78.07	**53.98**	72.97	78.40	**79.90**	**87.42**	**90.90**	80.71	85.11	57.88	**67.70**	68.85	69.96	50.24

the backbone of the detectors. All default hyperparameters of these detectors are followed and the batchsize is set at 2. We train the detectors on DOTA for 12 epochs and obtain detection accuracy by submitting the results on the testing set to the official evaluation server. For HRSC2016, we train the detectors for 36 epochs and report the mean accuracy (mAP) on the testing set. All experiments are implemented through OBBDetection[1] and conducted on a single V100 GPU.

2) Comparison With SOTA: The compared pretraining methods are the same as those in Section IV-B, and we report the detailed detection accuracy on DOTA using three rotated object detectors. For the sake of analysis, green indicates an increase and gray indicates a decrease. The results for two-stage Oriented R-CNN [23] and RoI Transformer [55] are shown in Tables III and IV, respectively. It can be seen that MGC-B performs better in predicting classes containing large-scale objects such as GTF, TC, and RA. The detection results of the one-stage detector S^2ANet [56] are reported in Table V. In addition to achieving the same high performance in TC and RA, MGC-L also has great prediction ability for small-scale and even dense objects such as BR, LV, and SH. Except using RoI Transformer detector, MGC pretrained models achieve promising accuracies and outperform INP. Compared to supervised RSP, a performance gap does exist,

[1]https://github.com/jbwang1997/OBBDetection

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TABLE VI
COMPARISON OF DETECTION RESULTS ON HRSC2016 USING EACH ROTATED OBJECT DETECTOR

Methods	Oriented. [23]	RoI Trans. [55]	S^2ANet [56]
	mAP		
Available Pre-trained Models			
INP [10]	**90.39**	90.21	89.61
RSP [11]	90.29	**90.34**	89.54
MoCo v2 [19]	90.03	88.74	89.19
SwAV [29]	90.08	**90.34**	**89.93**
SeCo [12]	88.85	89.36	78.21
Pre-trained on Fusion			
MoCo v2 [19]	88.98	89.51	86.62
SimCLR [17]	86.94	78.62	65.07
BYOL [18]	75.35	85.49	59.89
DetCo [30]	88.20	87.26	82.79
SwAV [29]	87.45	88.88	80.25
HCSC [34]	89.29	89.59	88.31
CARE [37]	89.63	89.85	88.62
MGC-B	89.96	90.16	88.92
MGC-L	89.86	90.06	89.20

but MGC achieves competitive results through unsupervised learning on a small-scale dataset alone. Table VI shows the detection results of all methods on HRSC2016, MGC-B, and MGC-L surpass all other contrastive learning-based pretraining methods and are comparable to the supervised INP.

Interestingly, MGC-B always outperforms MGC-L in the two-stage detectors, i.e., Oriented R-CNN and RoI Transformer. In contrast, for one-stage detector, S^2ANet MGC-L performs better than MGC-B. This is probably because the large range of contextual information in MGC-L inhibits the extraction of region proposals by the two-stage detectors, and MGC-B is adapted for subsequent classification of region proposals.

D. Semantic Segmentation

1) Experimental Setup: Three semantic segmentation frameworks, including UperNet [58], DeeplabV3+ [25], and PSPNet [59], are used to evaluate the performance of the pretrained model on the iSAID [26] and Potsdam[2] datasets.

1) iSAID is a large-scale instance segmentation dataset. Note that this dataset and DOTA share the same scenes, the difference is that iSAID is labeled with a semantic mask containing one background and 15 foregrounds class, including SH, ST, BD, TC, BC, GTF, BR, LV, SV, HC, SP, RA, SBF, PL, and HA. It consists of 2806 high-resolution images with pixel dimension ranging from 800 × 800 to 4000 × 13 000. The training, validation, and testing sets have 1411, 458, and 937 images, respectively. We crop all images to 896 × 896 patches with a stride of 512 and use only the validation set for evaluation since the testing set is unavailable.
2) Potsdam dataset is released by ISPRS Commission WG II/4. It contains 38 images with an average size of 6000 × 6000 pixels. These images cover a 3.42 km^2 area of Potsdam city and include six scenes, i.e., Impervious surface (IS), Building (BD), Low vegetation, Tree, Car, and Clutter. The training and testing sets have 24 and 14 images, respectively. Each image is cropped to 512 × 512 patches with a stride of 384. We exclude the clutter class from the dataset when calculating evaluation metrics.

[2]https://www.isprs.org/education/benchmarks/UrbanSemLab/2d-sem-label-potsdam.aspx

MMSEGMENTATION [60] is used to build all frameworks and experiments are conducted on one V100 GPU. All layers of the pretrained models are fine-tuned in an end-to-end manner. The fine-tuned models are used as the backbone in the three frameworks. For a fair comparison, all frameworks are trained 80k iterations using the original hyperparameters, and the batchsize is 8. The mean intersection over union (mIoU) for all classes (including background) and IoU per class on the iSAID validation set are reported. Overall accuracy (OA), mean F1 scores (mF1), and F1-score per class on the Potsdam testing set are also reported, respectively.

2) Comparison With SOTA: This section reports the detailed segmentation results on the two datasets in comparison to other pretrained models. MGC models perform superiorly on iSAID, where MGC-L even outperforms RSP by 0.37, 0.87, and 0.08 mIoU under UperNet, DeeplabV3+, and PSPNet frameworks, respectively. Specifically, Table VII shows the results using UperNet [58]. MGC-L achieves the best performance on BD, HC, and RA. For Deeplabv3+ [25], MGC models get the overall competitive results as shown in Table VIII, where we have the highest IoU on GTF, LV, SV, and BR. The segmentation results using PSPNet [59] are reported in Table IX, and MGC-L obtains the best for BD, GTF, and HC.

The results on Potsdam are, respectively, shown in Tables X–XII. MGC models outperform all other contrastive learning methods but generally underperform in comparison to two supervised pretraining models, INP and RSP.

E. Change Detection

1) Experimental Setup: BIT [5] is a unified change detection framework. Similarly, different pretrained models are used as the backbone of BIT for evaluation on the CDD [61] and LEVIR [27] datasets.

1) CDD is a benchmark that reflects real seasonal changes and includes seven pairs of 4725 × 2200 and four pairs of 1900 × 1000 pixels multisource RGB images. These images are collected from Google Earth with resolutions ranging from 0.03 to 1 m. we follow [62] and crop images into 256 × 256 patches. The training, validation, and testing sets contain 10 000, 3000, and 3000 patches, respectively.
2) LEVIR contains 637 pairs of 1024 × 1024 pixels images collected from Google Earth. These images cover 20 different areas in TX, USA, between 2002 and 2018 and contain 31 333 separate building change entities such as Villa, Apartment, Garage, and Warehouse. The training, validation, and testing sets contain 445, 64, and 128 pairs of images, respectively, and we crop images from LEVIR into 256 × 256 nonoverlapping patches, according to [5].

Default hyperparameters as those in [5] are used, and the models are trained 200 epochs through an stochastic gradient

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TABLE VII

SEGMENTATION RESULTS OF DIFFERENT PRETRAINED MODELS ON iSAID USING UPERNET [58]

Method	mIOU	IoU per class														
		Ship	ST	BD	TC	BC	GTF	Bridge	LV	SV	HC	SP	RA	SBF	Plane	Harbor
Available Pre-trained Models																
INP [10]	61.30	**64.83**	**71.20**	67.84	85.38	59.41	48.83	32.87	60.30	**49.03**	31.31	**47.71**	59.50	74.06	80.04	49.50
RSP [11]	61.28 (-0.02)	63.59	69.99	66.59	**88.78**	**60.14**	**50.88**	**38.21**	59.30	45.16	30.93	45.61	59.53	**75.83**	79.63	47.44
MoCo v2 [19]	40.87 (-20.43)	45.78	42.91	43.68	62.25	23.82	12.77	8.07	50.04	23.6	21.88	33.69	32.45	41.86	70.47	40.77
SwAV [29]	57.33 (-3.97)	63.47	70.24	59.32	69.37	54.36	45.47	35.67	54.15	46.00	24.49	43.30	54.85	70.93	81.35	45.45
SeCo [12]	57.45 (-3.85)	60.41	69.20	60.88	85.64	53.51	40.58	31.17	60.98	44.31	28.23	47.44	51.84	64.03	**80.46**	41.84
Pre-trained on Fusion																
MoCo v2 [19]	58.64 (-2.66)	60.81	63.19	61.20	86.11	56.68	44.50	32.81	59.06	45.23	27.91	47.55	54.40	70.60	80.44	49.00
SimCLR [17]	50.31 (-10.99)	55.74	59.36	54.81	82.29	31.12	41.19	14.99	53.67	35.30	20.37	42.92	39.45	52.41	76.35	46.34
BYOL [18]	50.76 (-10.54)	41.91	61.30	63.44	85.19	42.47	35.45	28.80	48.47	36.02	21.51	44.56	44.70	53.20	73.86	33.02
DetCo [30]	51.78 (-9.52)	54.46	61.02	51.33	84.29	37.92	31.81	22.99	52.75	38.97	22.07	43.96	39.91	65.34	79.23	43.72
SwAV [29]	55.76 (-5.54)	59.52	68.59	56.25	86.56	57.47	39.75	32.13	41.01	43.54	26.00	44.34	52.49	67.04	77.27	41.40
HCSC [34]	58.39 (-2.91)	61.88	68.10	62.57	87.27	58.63	40.18	31.19	57.20	42.31	25.53	45.71	55.21	70.82	78.91	**49.76**
CARE [37]	56.62 (-4.68)	59.31	68.56	58.33	85.64	50.88	39.99	28.12	58.25	44.87	25.86	45.37	52.75	64.17	77.71	47.31
MGC-B	61.05 (-0.25)	63.54	70.42	66.18	87.18	59.31	48.65	36.16	**61.20**	47.47	31.84	46.92	56.12	74.92	79.93	47.77
MGC-L	**61.65** (+0.35)	64.04	71.12	**69.14**	87.51	59.48	48.57	34.01	59.44	47.02	**32.47**	47.64	**63.25**	74.58	79.97	49.27

TABLE VIII

SEGMENTATION RESULTS OF DIFFERENT PRETRAINED MODELS ON iSAID USING DEEPLABV3+ [25]

Method	mIOU	IoU per class														
		Ship	ST	BD	TC	BC	GTF	Bridge	LV	SV	HC	SP	RA	SBF	Plane	Harbor
Available Pre-trained Models																
INP [10]	58.75	62.52	65.01	**70.84**	86.18	54.87	48.56	34.96	58.62	44.94	29.15	47.16	50.87	71.58	66.05	**49.91**
RSP [11]	58.96 (+0.21)	60.85	66.85	63.18	**87.39**	**60.03**	43.54	33.95	58.11	43.38	**29.77**	**48.79**	51.42	**75.55**	76.45	45.14
MoCo v2 [19]	43.80 (-14.95)	50.32	53.50	53.91	58.56	24.42	31.18	10.60	48.24	32.52	18.65	38.05	24.25	45.74	68.91	42.10
SwAV [29]	56.19 (-2.56)	**62.76**	67.62	67.98	64.43	54.20	44.57	36.42	54.95	41.47	24.04	42.83	53.10	65.12	73.05	46.83
SeCo [12]	57.17 (-1.58)	59.55	60.60	68.69	86.38	52.69	40.25	32.20	59.83	42.72	28.68	46.88	45.57	66.31	**79.88**	45.62
Pre-trained on Fusion																
MoCo v2 [19]	57.94 (-0.81)	61.22	61.09	68.13	84.61	56.51	41.72	33.43	59.23	43.93	26.86	46.00	50.35	71.53	76.63	46.95
SimCLR [17]	48.28 (-10.47)	50.72	55.23	57.45	74.57	25.11	33.24	14.77	55.47	37.47	19.54	44.22	38.06	53.90	73.61	40.48
BYOL [18]	44.96 (-13.79)	40.93	52.57	47.11	58.98	30.17	38.62	33.49	44.33	33.60	17.12	44.63	30.45	49.08	68.22	31.77
DetCo [30]	51.69 (-7.06)	53.78	58.82	57.06	84.06	39.85	34.56	21.46	55.44	38.60	21.65	46.49	34.91	67.25	74.29	40.14
SwAV [29]	54.26 (-4.49)	57.79	60.91	66.97	81.92	49.03	38.78	36.37	40.11	34.24	24.04	45.62	52.32	66.80	75.23	39.38
HCSC [34]	58.11 (-0.64)	58.74	64.69	67.16	84.71	54.54	42.35	31.84	60.23	41.98	26.37	48.09	**58.47**	69.10	76.16	46.42
CARE [37]	57.23 (-1.52)	59.49	**68.29**	63.57	82.96	55.29	35.90	34.20	60.93	42.68	25.99	47.58	53.37	59.55	78.81	48.41
MGC-B	59.64 (+0.89)	62.16	62.29	68.50	86.20	56.95	**49.08**	35.32	**61.07**	**45.98**	29.50	46.85	56.26	71.66	78.50	45.11
MGC-L	**59.83** (+1.08)	61.70	67.80	70.49	86.00	55.94	47.39	**36.54**	60.87	45.71	29.71	47.58	55.35	70.29	75.62	47.31

TABLE IX

SEGMENTATION RESULTS OF DIFFERENT PRETRAINED MODELS ON iSAID USING PSPNET [59]

Method	mIOU	IoU per class														
		Ship	ST	BD	TC	BC	GTF	Bridge	LV	SV	HC	SP	RA	SBF	Plane	Harbor
Available Pre-trained Models																
INP [10]	51.26	**47.82**	**58.34**	67.39	82.65	56.83	48.42	33.14	**46.23**	20.79	8.27	36.68	57.43	74.61	56.72	26.34
RSP [11]	51.28 (+0.02)	47.42	57.72	64.48	**83.22**	**59.55**	47.70	32.15	45.37	19.05	8.44	37.94	**59.86**	74.99	58.55	25.41
MoCo v2 [19]	31.23 (-20.03)	27.20	40.25	36.92	52.89	22.27	24.73	7.24	39.66	7.23	1.87	27.43	2.54	46.61	47.86	15.09
SwAV [29]	48.16 (-3.10)	46.55	50.34	67.50	65.23	54.46	48.01	**33.58**	40.12	20.19	4.77	32.77	52.98	**75.26**	54.08	24.83
SeCo [12]	48.32 (-2.94)	41.67	56.30	66.40	80.70	51.81	44.86	30.71	44.25	18.15	7.20	36.56	40.71	69.19	57.73	**28.49**
Pre-trained on Fusion																
MoCo v2 [19]	49.21 (-2.05)	45.07	54.21	63.27	80.83	57.93	49.62	29.27	43.07	19.00	8.15	34.85	51.60	73.00	55.24	23.70
SimCLR [17]	43.13 (-8.13)	39.60	43.92	53.26	76.06	32.72	38.57	19.53	43.11	13.80	4.02	37.33	41.64	66.33	57.76	24.10
BYOL [18]	41.12 (-10.14)	34.90	44.13	63.25	74.78	23.25	37.90	28.58	27.67	10.62	3.44	30.51	44.01	66.57	48.70	21.40
DetCo [30]	42.87 (-8.39)	39.30	46.79	50.83	75.34	38.55	38.72	17.65	37.96	13.76	3.69	**40.96**	38.64	67.70	53.98	23.76
SwAV [29]	48.07 (-3.19)	42.03	52.31	66.11	76.54	56.18	44.46	32.26	36.63	15.72	6.02	33.78	53.89	70.77	58.47	25.44
HCSC [34]	49.98 (-1.28)	45.10	54.33	64.99	82.78	58.68	46.95	30.10	44.70	20.23	6.83	35.28	51.62	72.92	**59.71**	26.92
CARE [37]	46.34 (-4.92)	38.27	51.57	57.47	78.57	48.80	47.36	27.90	41.19	15.31	6.27	39.89	44.35	70.79	52.61	22.81
MGC-B	50.68 (-0.58)	45.63	57.26	67.04	82.05	55.98	51.24	30.48	46.15	**23.21**	8.67	37.71	52.52	72.20	56.39	25.94
MGC-L	**51.36** (+0.10)	45.36	55.29	**68.73**	81.52	58.65	**52.19**	31.36	45.83	21.35	**9.23**	38.45	58.87	72.96	55.43	27.93

descent with momentum (SGDM) optimizer with a momentum of 0.99 and a weight decay of 0.0005. The initial learning rate is set at 0.001, and a linear decay strategy is used. The batchsize is 32 for CDD and 4 for LEVIR. Similarly, experiments are conducted on one V100 GPU. F1 scores on the two testing sets are reported.

2) Comparison With SOTA: The F1 scores of all methods fine-tuned on the two datasets are shown in Table XIII. Change detection aims to match pixels at the same position of a pair of dual-temporal images, so MGC-B with enhanced local recognition outperforms MGC-L. Higher F1 scores than all other contrastive learning methods are obtained using the MGC models. MGC-B even outperforms the supervised RSP on CDD.

F. Ablation Study

The ablation study is based on MGC-B and all results we report are obtained by fine-tuning the model with 200 epochs. Oriented R-CNN [23] is used as the rotated object detector, and UperNet [58] as the semantic segmentation framework.

1) Numbers of CycleMLP: n is determined according to the experimental results in Table XIV. The best accuracy on the

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TABLE X

SEGMENTATION RESULTS OF DIFFERENT PRETRAINED MODELS ON POTSDAM USING UPERNET [58]

Method	OA	mF1	F1 score per class				
			IS	BD	LV	Tree	Car
Available Pre-trained Models							
INP [10]	**91.22**	**90.71**	93.01	**96.78**	**86.39**	**86.10**	**91.25**
RSP [11]	91.02	90.42	92.94	96.75	86.13	85.61	90.66
MoCo v2 [19]	90.58	90.18	92.56	96.39	85.81	85.25	90.90
SwAV [29]	90.67	90.21	92.45	96.18	85.84	85.55	91.01
SeCo [12]	90.45	89.83	91.37	95.85	85.77	85.46	90.69
Pre-trained on Fusion							
MoCo v2 [19]	90.85	90.24	92.51	96.39	85.86	85.56	90.86
SimCLR [17]	90.41	89.82	92.44	95.72	85.42	85.29	90.21
BYOL [18]	89.71	89.17	91.53	95.38	84.65	84.74	89.54
DetCo [30]	90.66	90.12	**93.22**	96.09	85.54	85.36	90.38
SwAV [29]	90.54	90.08	92.32	96.32	85.55	85.70	90.48
HCSC [34]	90.74	90.17	92.52	96.12	85.67	85.89	90.61
CARE [37]	90.91	90.32	93.10	96.22	85.87	85.57	90.81
MGC-B	90.92	90.38	92.82	96.62	86.06	85.48	90.95
MGC-L	90.90	90.36	92.83	96.59	85.98	85.54	90.88

TABLE XI

SEGMENTATION RESULTS OF DIFFERENT PRETRAINED MODELS ON POTSDAM USING DEEPLABV3+ [25]

Method	OA	mF1	F1 score per class				
			IS	BD	LV	Tree	Car
Available Pre-trained Models							
INP [10]	**91.03**	**90.45**	93.03	96.40	85.79	**86.28**	**90.75**
RSP [11]	90.90	90.37	**93.30**	**96.45**	85.76	85.86	90.46
MoCo v2 [19]	90.48	90.03	92.34	96.05	85.63	85.52	90.62
SwAV [29]	90.58	90.11	92.45	96.27	85.81	85.44	90.58
SeCo [12]	90.33	89.59	91.50	95.76	85.05	85.62	90.00
Pre-trained on Fusion							
MoCo v2 [19]	90.38	89.67	91.34	95.95	85.37	85.57	90.10
SimCLR [17]	89.85	89.07	90.68	95.26	84.83	85.35	89.21
BYOL [18]	89.69	88.92	91.08	95.07	84.45	85.08	88.94
DetCo [30]	90.21	89.57	91.82	95.44	84.93	85.82	89.85
SwAV [29]	90.39	89.69	91.74	96.04	85.20	85.53	89.96
HCSC [34]	89.90	89.30	90.96	95.30	84.83	85.34	90.05
CARE [37]	90.44	90.03	92.39	96.01	85.66	85.74	90.34
MGC-B	90.77	90.22	92.60	96.43	**85.85**	85.57	90.67
MGC-L	90.45	89.92	92.32	95.87	85.42	85.63	90.34

TABLE XII

SEGMENTATION RESULTS OF DIFFERENT PRETRAINED MODELS ON POTSDAM USING PSPNET [59]

Method	OA	mF1	F1 score per class				
			IS	BD	LV	Tree	Car
Available Pre-trained Models							
INP [10]	**90.35**	**88.41**	**91.96**	96.45	**85.53**	**85.67**	82.44
RSP [11]	90.04	88.18	91.69	96.28	85.19	85.19	**82.55**
MoCo v2 [19]	89.27	87.89	91.48	95.98	85.03	85.30	81.65
SwAV [29]	89.65	88.08	91.59	96.14	85.12	85.36	82.21
SeCo [12]	89.94	87.79	90.40	96.35	85.22	85.11	81.91
Pre-trained on Fusion							
MoCo v2 [19]	89.74	87.67	90.63	96.09	85.02	84.80	81.86
SimCLR [17]	89.45	87.08	88.83	95.78	84.67	84.69	81.46
BYOL [18]	88.86	86.22	86.95	95.55	83.83	84.04	80.77
DetCo [30]	89.74	87.37	89.70	96.05	84.82	84.71	81.58
SwAV [29]	89.48	87.41	89.74	95.95	84.48	85.17	81.75
HCSC [34]	89.71	87.81	91.18	96.03	84.98	85.20	81.68
CARE [37]	89.76	87.90	91.08	**96.60**	85.16	84.66	82.02
MGC-B	90.06	88.13	91.75	96.15	85.20	85.33	82.24
MGC-L	90.05	88.07	91.69	96.22	85.21	85.32	81.92

largest dataset MLRSNet is obtained when $n = 6$. Therefore, we set the numbers of CycleMLP at 6 for the experiments on NWPU45 and AID as well.

2) Weight of Losses: In Table XIV, the best performance on MLRSNet and AID is obtained when $\lambda_C = 1$. The effect of adjusting λ_P is shown in Fig. 4(a), and the model achieves the best classification results on all datasets when $\lambda_P = 200$. As shown in Fig. 4(b), when $\lambda_A = 50$, the model achieves the best accuracies on MLRSNet and NWPU45 and good results on AID.

TABLE XIII

CHANGE DETECTION RESULTS OF DIFFERENT PRETRAINED MODELS UNDER THE BIT [5] FRAMEWORK

Methods	CDD	LEVIR
	F1 Score	
Available Pre-trained Models		
INP [10]	**96.18**	**87.21**
RSP [11]	93.99	86.95
MoCo v2 [19]	94.95	85.81
SwAV [29]	95.17	85.92
SeCo [12]	93.75	84.35
Pre-trained on Fusion		
MoCo v2 [19]	94.97	85.89
SimCLR [17]	89.07	80.52
BYOL [18]	93.14	79.70
DetCo [30]	95.03	85.87
SwAV [29]	91.91	82.92
HCSC [34]	94.11	85.47
CARE [37]	95.14	86.05
MGC-B	95.20	86.14
MGC-L	94.95	85.58

TABLE XIV

EFFECT OF DIFFERENT λ_C AND n ON MGC SCENE CLASSIFICATION

n	2	3	4	5	6	7
MLRSNet	95.83	95.88	95.73	95.88	**95.95**	95.57
NWPU45	**92.63**	92.43	92.33	92.41	92.50	92.01
AID	94.12	94.08	**94.28**	93.78	94.14	93.86
λ_C	0	0.1	0.5	1	2	5
MLRSNet	95.88	95.82	95.75	**95.97**	95.75	95.46
NWPU45	**92.70**	92.45	92.04	92.51	92.14	91.71
AID	94.06	94.02	94.06	**94.08**	93.88	93.96

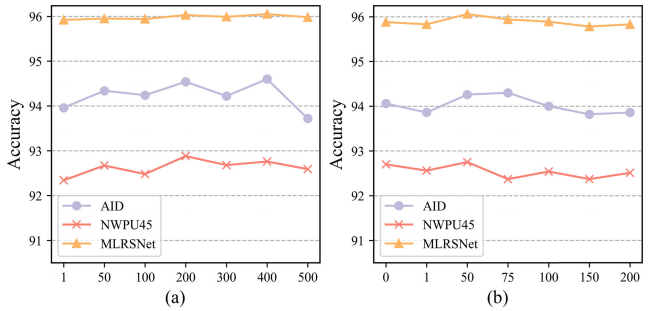


Fig. 4. Effect of different (a) λ_P and (b) λ_A on MGC scene classification accuracy.

3) Effectiveness of Each Loss: Contributions of different losses in pretraining are reported in Table XV. Compared with $\mathcal{L}_{\text{con}}^{\text{same}}$ alone, adding $\mathcal{L}_{\text{con}}^{\text{corss}}$ increases the performance of the pretrained model significantly. Although classification accuracies in the third and fourth rows of Table XV decrease, improvements have been achieved on other tasks. Furthermore, when using both losses to pretrain the model, MGC achieves higher mAP and mIoU on DOTA and iSAID, 0.62 and 0.58 higher than the case when only contrastive losses are involved, respectively. The above results demonstrate the effectiveness of the proposed MGC method.

4) Step Size: Table XVI reports the ablation results with a different step size s. The MLP network in $E_{\cdot}^{m}$ with $s = 1$ is formed by general MLPs, and the transferred model is not friendly to dense prediction tasks at this time. When $s = 3$, the discriminative information extracted from small-scale scenes

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TABLE XV
EFFECT OF DIFFERENT LOSSES ON MGC PERFORMANCES OF MULTIPLE TASKS

$\mathcal{L}_{con}^{\text{same}}$	$\mathcal{L}_{con}^{\text{cross}}$	$\mathcal{L}_{kl}^{\text{PA}}$	$\mathcal{L}_{kl}^{\text{AA}}$	MLRSNet	NWPU45	AID	DOTA	iSAID
				Accuracy			mAP	mIoU
✓				95.88	**92.70**	94.06	74.14	59.78
✓	✓			95.97 (+0.09)	92.51 (-0.19)	94.08 (+0.02)	74.78 (+0.64)	60.26 (+0.48)
✓	✓	✓		**96.08** (+0.20)	92.32 (-0.38)	93.88 (-0.18)	75.08 (+0.94)	60.50 (+0.72)
✓	✓		✓	95.94 (+0.06)	92.23 (-0.47)	94.06 (-0.00)	75.00 (+0.86)	60.57 (+0.79)
✓	✓	✓	✓	95.95 (+0.07)	92.50 (-0.20)	**94.14** (+0.08)	**75.40** (+1.26)	**60.84** (+1.06)

TABLE XVI
EFFECT OF DIFFERENT s ON MGC PERFORMANCES OF MULTIPLE TASKS

s	MLRSNet	NWPU45	AID	DOTA	iSAID
	Accuracy			mAP	mIoU
1	95.86	92.38	94.27	75.28	60.71
3	**95.95**	**92.50**	94.14	**75.40**	60.84
5	95.74	92.30	94.00	75.16	60.66
7	95.62	91.87	93.88	75.03	60.33
9	95.55	91.96	93.70	74.85	61.08
11	95.83	92.15	**94.32**	74.96	**61.23**
13	95.80	92.07	93.62	74.71	61.05

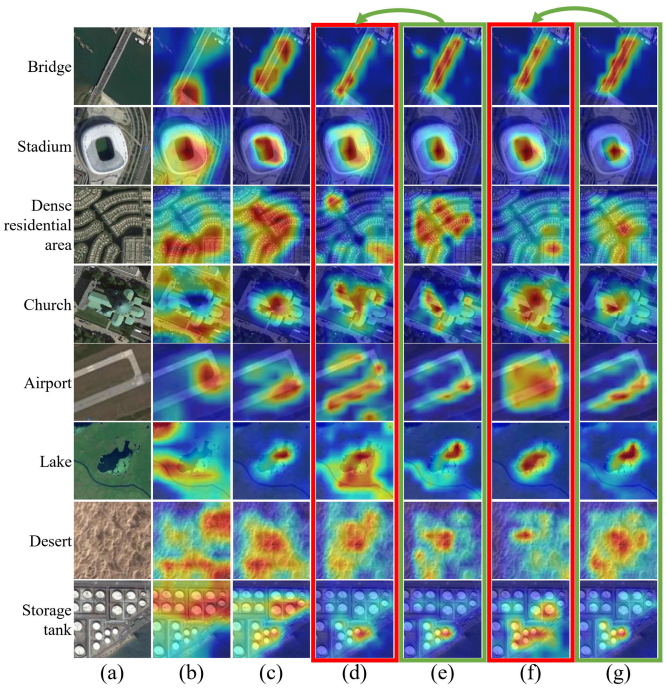


Fig. 5. Attention maps of different models on various scenes: (a) image; (b) INP; (c) MoCo v2 [19]; (d) CNN backbone of MGC-B; (e) MLP backbone of MGC-B; (f) CNN backbone of MGC-L; and (g) MLP backbone of MGC-L. MGC's MLP backbones (green box) effectively guide the CNN backbones (red box) to focus on correct foreground regions.

facilitates the recognition of $E_{:}^{c}$, and the highest performance is obtained except on AID and iSAID. $E_{:}^{m}$ continuously extends the range of feature aggregation with increasing s, thus gradually improves the locating ability of $E_{:}^{c}$. Until $s = 11$, the model's performance reaches its top because the foreground regions are precisely located. Therefore, the two settings of $s = 3$ and $s = 11$ are used in the experiments.

5) Visualization: Fig. 5 provides visualizations of attention from different models on diverse scenes using GradCAM++ [15]. It can be seen that CNN backbones (red box), guided by MLP backbones (green box), focus their attention on the correct foreground regions, regardless of whether they are dense or sparse. The attention of MGC-L is more discriminative compared to MGC-B, which explains why the transferred CNN backbone of MGC-L exhibits stronger locating ability. For example, MGC-L reduces redundant attention in pure textured scenes such as Desert; the texture information learned by the CNN backbone of MGC-L from small regions is highly discriminative. In comparison, the MoCo v2 pretrained model displays incomplete attention toward the object with a large aspect ratio (BR and airport) or the regional object (church and lake), and it exhibits redundant attention toward dense objects (dense residential area). INP generates incorrect attention due to the domain gap.

V. CONCLUSION

This article proposes an MLP-guided CNN pretraining method, MGC. Its effectiveness attributes to the proposed MLP branches to guide the CNN branches to overcome the locality issue of shallow layers and focus on the correct foreground regions. MGC pretrained models achieve the state-of-the-art scene classification accuracy while providing competitive performance on rotated object detection, semantic segmentation, and change detection. The proposed MGC offers a useful pretraining method for the RSI community, and its framework may also be useful for other special types of images such as hyperspectral or SAR images.

REFERENCES

[1] G. Schumann, G. Brakenridge, A. Kettner, R. Kashif, and E. Niebuhr, "Assisting flood disaster response with Earth observation data and products: A critical assessment," *Remote Sens.*, vol. 10, no. 8, p. 1230, Aug. 2018.
[2] D. Rolnick et al., "Tackling climate change with machine learning," *ACM Comput. Surv.*, vol. 55, no. 2, pp. 1–96, Feb. 2022.
[3] D. J. Mulla, "Twenty five years of remote sensing in precision agriculture: Key advances and remaining knowledge gaps," *Biosyst. Eng.*, vol. 114, no. 4, pp. 358–371, Apr. 2013.
[4] C. Ippoliti et al., "Defining ecological regions in Italy based on a multivariate clustering approach: A first step towards a targeted vector borne disease surveillance," *PLoS ONE*, vol. 14, no. 7, Jul. 2019, Art. no. e0219072.
[5] S.-B. Chen, Q.-S. Wei, W.-Z. Wang, J. Tang, B. Luo, and Z.-Y. Wang, "Remote sensing scene classification via multi-branch local attention network," *IEEE Trans. Image Process.*, vol. 31, pp. 99–109, 2022.
[6] Y. Sun, L. Lei, D. Guan, and G. Kuang, "Iterative robust graph for unsupervised change detection of heterogeneous remote sensing images," *IEEE Trans. Image Process.*, vol. 30, pp. 6277–6291, 2021.
[7] R. Xu, C. Wang, J. Zhang, S. Xu, W. Meng, and X. Zhang, "RSSFormer: Foreground saliency enhancement for remote sensing land-cover segmentation," *IEEE Trans. Image Process.*, vol. 32, pp. 1052–1064, 2023.
[8] Z. Xue, X. Tan, X. Yu, B. Liu, A. Yu, and P. Zhang, "Deep hierarchical vision transformer for hyperspectral and LiDAR data classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3095–3110, 2022.

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[9] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[10] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.

[11] D. Wang, J. Zhang, B. Du, G.-S. Xia, and D. Tao, "An empirical study of remote sensing pretraining," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5608020.

[12] O. Manas, A. Lacoste, X. Giro-i-Nieto, D. Vazquez, and P. Rodriguez, "Seasonal contrast: Unsupervised pre-training from uncurated remote sensing data," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9414–9423.

[13] Y. Cong et al., "SatMAE: Pre-training transformers for temporal and multi-spectral satellite imagery," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 197–211.

[14] X. Sun et al., "RingMo: A remote sensing foundation model with masked image modeling," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5612822.

[15] A. Chattopadhay, A. Sarkar, P. Howlader, and V. N. Balasubramanian, "Grad-CAM++: Generalized gradient-based visual explanations for deep convolutional networks," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2018, pp. 839–847.

[16] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9729–9738.

[17] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1597–1607.

[18] J.-B. Grill et al., "Bootstrap your own latent-a new approach to self-supervised learning," in *Proc. 34th Int. Conf. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 21271–21284.

[19] X. Chen, H. Fan, R. Girshick, and K. He, "Improved baselines with momentum contrastive learning," 2020, *arXiv:2003.04297*.

[20] G.-S. Xia et al., "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.

[21] X. Qi et al., "MLRSNet: A multi-label high spatial resolution remote sensing dataset for semantic scene understanding," *ISPRS J. Photogramm. Remote Sens.*, vol. 169, pp. 337–350, Nov. 2020.

[22] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.

[23] X. Xie, G. Cheng, J. Wang, X. Yao, and J. Han, "Oriented R-CNN for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3520–3529.

[24] G.-S. Xia et al., "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.

[25] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.

[26] S. W. Zamir et al., "iSAID: A large-scale dataset for instance segmentation in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, Jul. 2019, pp. 28–37.

[27] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, 2020.

[28] X. Chen and K. He, "Exploring simple Siamese representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 15750–15758.

[29] M. Caron, I. Misra, J. Mairal, P. Goyal, P. Bojanowski, and A. Joulin, "Unsupervised learning of visual features by contrasting cluster assignments," in *Proc. NIPS*, Dec. 2020, pp. 9912–9924.

[30] E. Xie et al., "DetCo: Unsupervised contrastive learning for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 8392–8401.

[31] J. Robinson, C.-Y. Chuang, S. Sra, and S. Jegelka, "Contrastive learning with hard negative samples," 2020, *arXiv:2010.04592*.

[32] R. Zhu, B. Zhao, J. Liu, Z. Sun, and C. W. Chen, "Improving contrastive learning by visualizing feature transformation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10306–10315.

[33] J. Li, P. Zhou, C. Xiong, and S. C. H. Hoi, "Prototypical contrastive learning of unsupervised representations," 2020, *arXiv:2005.04966*.

[34] Y. Guo et al., "HCSC: Hierarchical contrastive selective coding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 9696–9705.

[35] R. Devon Hjelm et al., "Learning deep representations by mutual information estimation and maximization," 2018, *arXiv:1808.06670*.

[36] H. Wang, X. Guo, Z. Deng, and Y. Lu, "Rethinking minimal sufficient representation in contrastive learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 16041–16050.

[37] C. Ge, Y. Liang, Y. Song, J. Jiao, J. Wang, and P. Luo, "Revitalizing CNN attention via transformers in self-supervised visual representation learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 4193–4206.

[38] H. Cheng et al., "Unsupervised visual representation learning via multi-dimensional relationship alignment," *IEEE Trans. Image Process.*, vol. 32, pp. 1613–1626, 2023.

[39] P. Khosla et al., "Supervised contrastive learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 18661–18673.

[40] H. Bao, L. Dong, S. Piao, and F. Wei, "BEiT: BERT pre-training of image transformers," 2021, *arXiv:2106.08254*.

[41] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 16000–16009.

[42] P. Akiva, M. Purri, and M. Leotta, "Self-supervised material and texture representation learning for remote sensing tasks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 8203–8215.

[43] K. Ayush et al., "Geography-aware self-supervised learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10181–10190.

[44] W. Li, K. Chen, H. Chen, and Z. Shi, "Geographical knowledge-driven representation learning for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5405516.

[45] J. Kang, R. Fernandez-Beltran, P. Duan, S. Liu, and A. J. Plaza, "Deep unsupervised embedding for remotely sensed images based on spatially augmented momentum contrast," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2598–2610, Mar. 2021.

[46] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[47] G. Christie, N. Fendley, J. Wilson, and R. Mukherjee, "Functional map of the world," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6172–6180.

[48] C. J. Reed et al., "Scale-MAE: A scale-aware masked autoencoder for multiscale geospatial representation learning," 2022, *arXiv:2212.14532*.

[49] Y. Long et al., "On creating benchmark dataset for aerial image interpretation: Reviews, guidances, and million-AID," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4205–4230, 2021.

[50] S. Chen, E. Xie, C. Ge, R. Chen, D. Liang, and P. Luo, "CycleMLP: A MLP-like architecture for dense prediction," 2021, *arXiv:2107.10224*.

[51] A. van den Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," 2018, *arXiv:1807.03748*.

[52] Z. Wu, Y. Xiong, S. X. Yu, and D. Lin, "Unsupervised feature learning via non-parametric instance discrimination," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3733–3742.

[53] P. Helber, B. Bischke, A. Dengel, and D. Borth, "EuroSAT: A novel dataset and deep learning benchmark for land use and land cover classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 7, pp. 2217–2226, Jul. 2019.

[54] M. Neumann, A. Susano Pinto, X. Zhai, and N. Houlsby, "In-domain representation learning for remote sensing," 2019, *arXiv:1911.06721*.

[55] J. Ding, N. Xue, Y. Long, G.-S. Xia, and Q. Lu, "Learning RoI transformer for oriented object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2849–2858.

[56] J. Han, J. Ding, J. Li, and G.-S. Xia, "Align deep features for oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2021.

[57] Z. Liu, L. Yuan, L. Weng, and Y. Yang, "A high resolution optical satellite image dataset for ship recognition and some new baselines," in *Proc. Int. Conf. Pattern Recognit. Appl. Methods*, vol. 2, 2017, pp. 324–331.

[58] T. Xiao, Y. Liu, B. Zhou, Y. Jiang, and J. Sun, "Unified perceptual parsing for scene understanding," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 418–434.

[59] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.

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[60] M. Contributors. (2020). *MMSegmentation: OpenMMLab Semantic Seg-Mentation Toolbox and Benchmark*. [Online]. Available: https://github.com/open-mmlab/mmsegmentation

[61] M. A. Lebedev, Y. V. Vizilter, O. V. Vygolov, V. A. Knyaz, and A. Y. Rubis, "Change detection in remote sensing images using conditional adversarial networks," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 42, no. 2, pp. 565–571, May 2018.

[62] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2018.



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